SPARK ACADEMY 2026

TRAIN FOR CHANGE, FROM SCIENCE TO PRACTICE

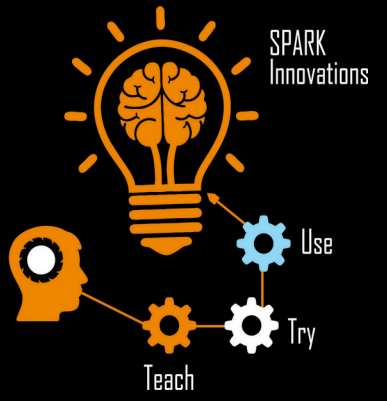
THE SPRINT AI TRAINING FOR AFRICAN MEDICAL
Imaging Knowledge Translation (SPARK) Academy

IN DEEP LEARNING & MEDICAL IMAGING

?What is AI

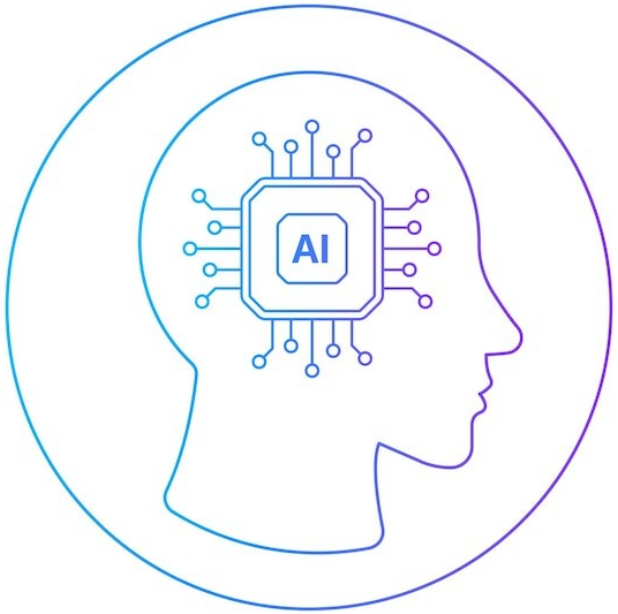
BrainHack 2023

Train for Change, from Science to Practice



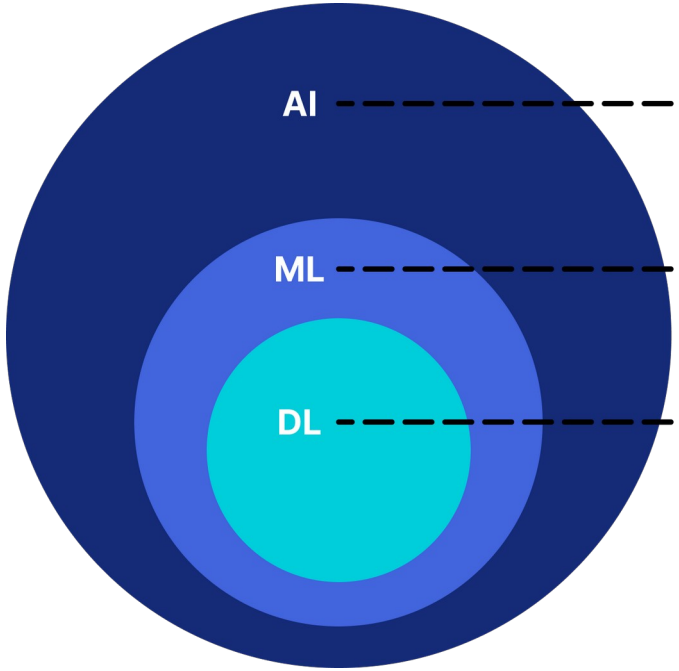
?What is AI

Sub-field of **computer science**. Artificial Intelligence (AI) studies how we can make the computer do things .that humans can still do better at the moment



https://www.freepik.com/vectors/artificial-intelligence-png

That sounds hard …. Let’s break it down



Artificial Intelligence
Engineering of making intelligent machines and programs.

Machine Learning
Ability to learn without being explicitly programmed.

Deep Learning
Learning based on deep neural networks.

neurosnap.ai

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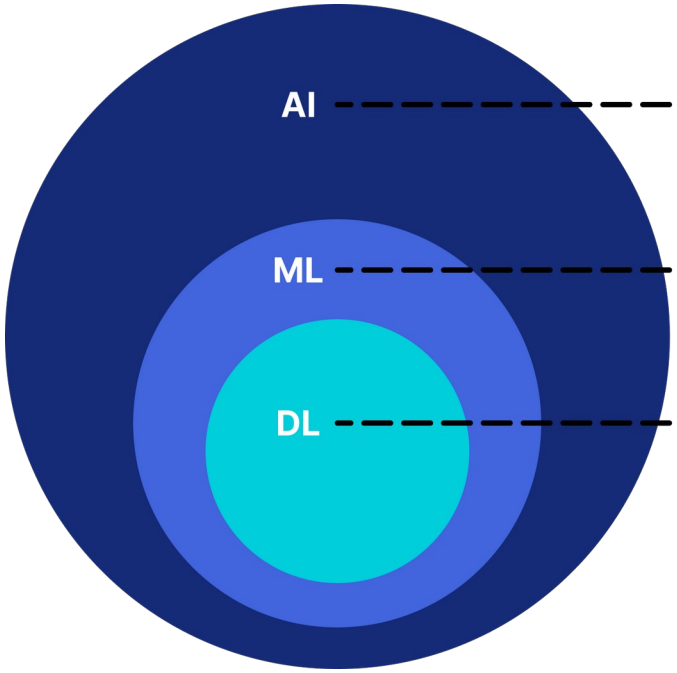

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AI

When AI started it was about **searching** for a solution
.that answers the problem

1. ∀x (Cat(x) → Mammal(x)) (All cats are mammals)
2. ∀x (Mammal(x) → Animal(x)) (All mammals are animals)
3. Cat(Tom) (Tom is a cat)

From these statements, we can infer:

1. Mammal(Tom) (Since Tom is a cat, and all cats are mammals)
2. Animal(Tom) (Since Tom is a mammal, and all mammals are animals)

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ML/DL

ML tries to learn the pattern. In ML there are 3 main pillars

Supervised

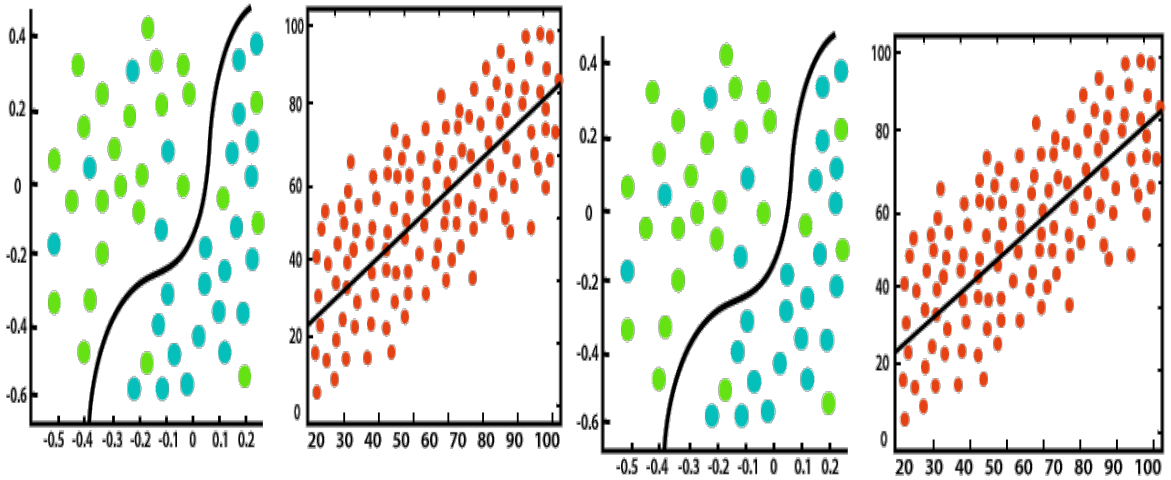
Un-supervised

Reinforcement

https://www.freepik.com/vectors/artificial-intelligence-png

Supervised

Try to give the model A LOT of examples (input/output) and it .learns from them



classification

Regressio
n

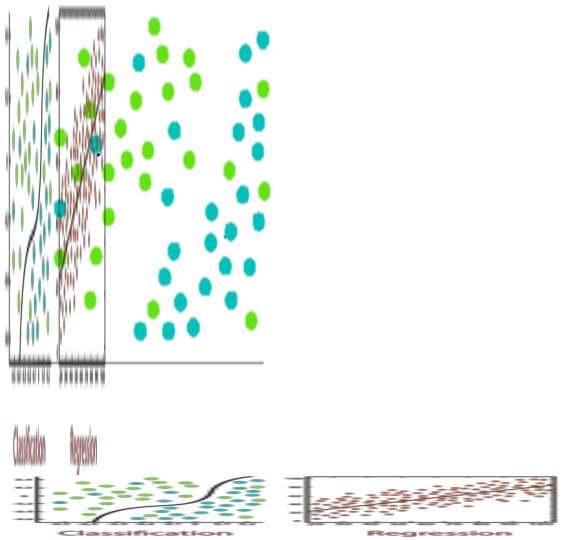
Segmentatio
n

Supervised

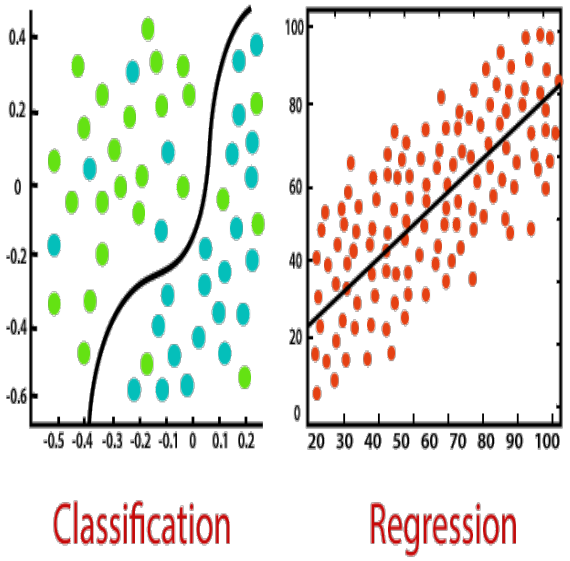
X-axis	Y-axis	label
0.4	-0.4	●
0	0	●
-0.4	0.1	●
. . .		

Object location

Is it green or ?blue



Object location



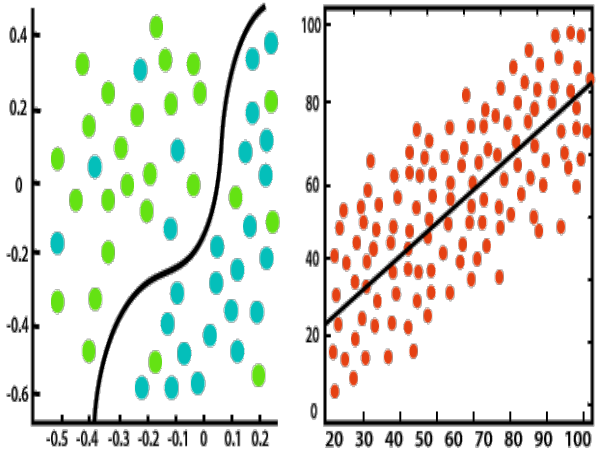
classification

Trained model

Supervised

What is the
?temperature

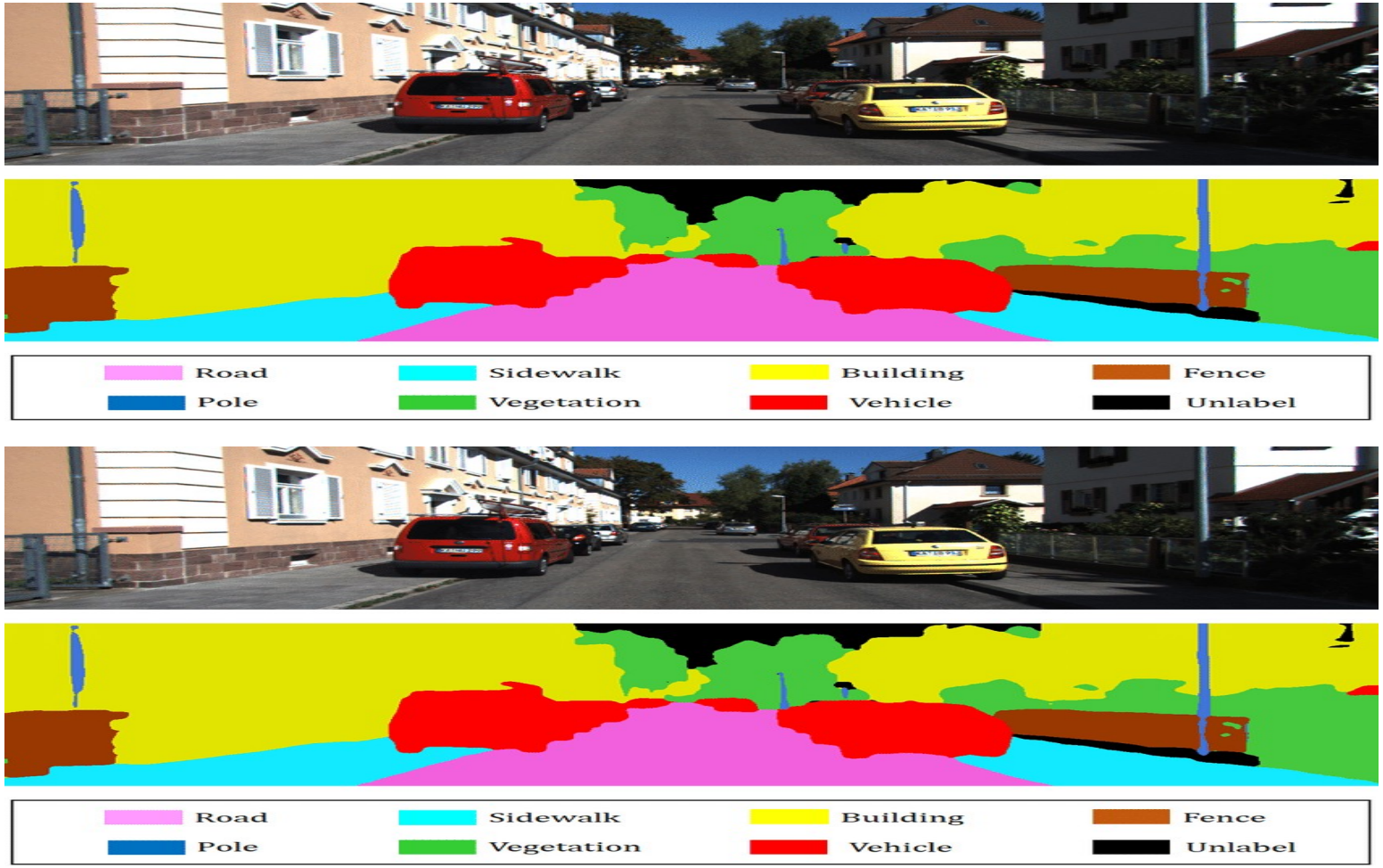
Distance	temperature
40	30
100	100
50	55.3
. . .	



Classification Regression

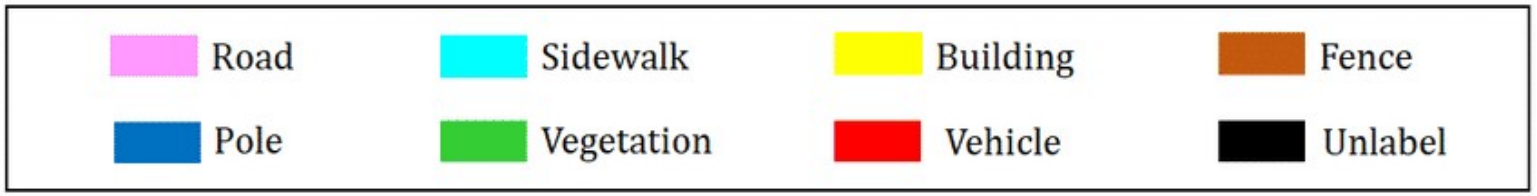
Regressio
n

Supervised



Segmentatio
n

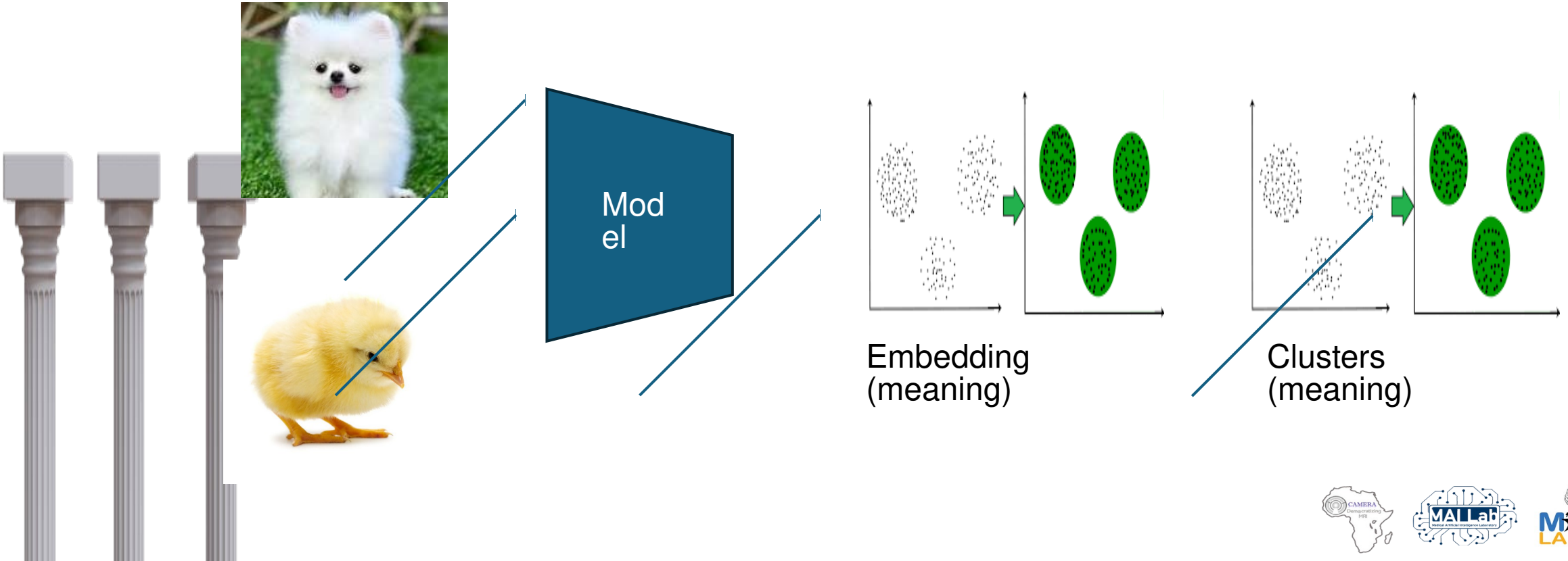
Supervised



Segmentation + classification

Un-supervised

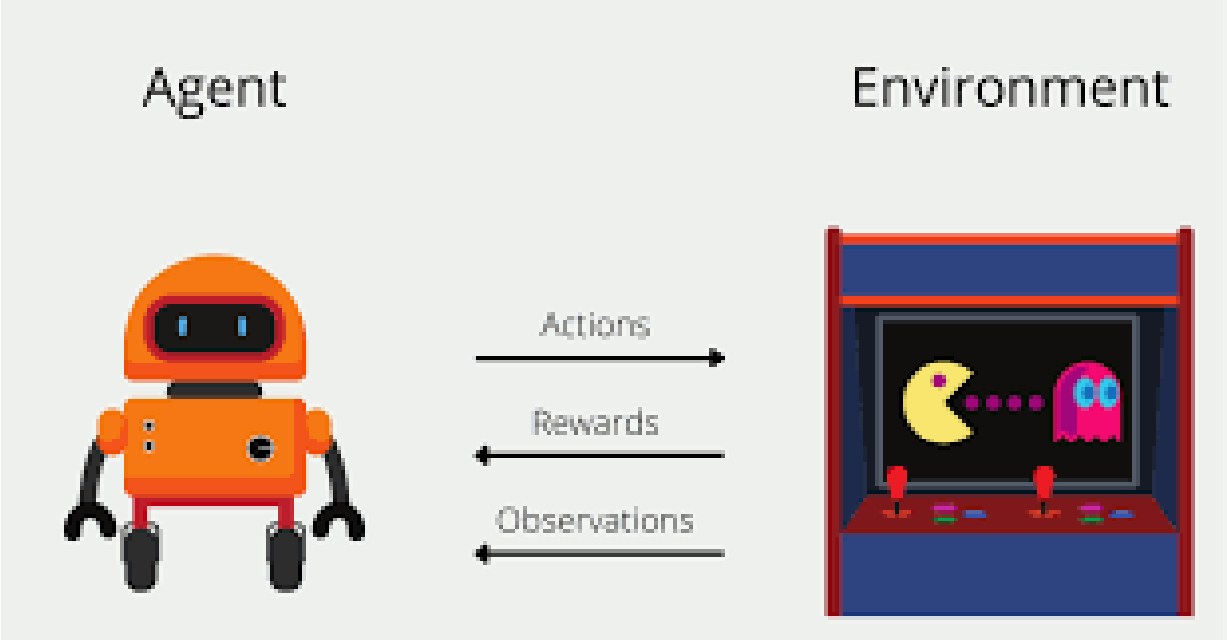
Try to give the model A LOT of examples but without label
.and it learns from them



Reinforcement

The model plays in a simulation, it gets a reward for every good thing it makes and gets punishment for every bad thing

am like babies but 10000000x dumber .....



There are a lot more to learn

General

- Attention — 132 methods, 47294 papers with code
- Activation Functions — 86 methods, 37907 papers with code
- Attention Mechanisms — 86 methods, 27256 papers with code
- Stochastic Optimization — 72 methods, 28246 papers with code
- Regularization — 62 methods, 62707 papers with code

See all 134 categories

Computer Vision

- Convolutional Neural Networks — 126 methods, 6461 papers with code
- Generative Models — 101 methods, 10165 papers with code
- Image Model Blocks — 93 methods, 8067 papers with code
- Object Detection Models — 73 methods, 2935 papers with code
- Image Feature Extractors — 63 methods, 188 papers with code

See all 126 categories

Natural Language Processing

- Language Models — 141 methods, 48595 papers with code
- Transformers — 95 methods, 41115 papers with code
- Word Embeddings — 21 methods, 916 papers with code
- Sentence Embeddings — 8 methods, 216 papers with code
- Attention Patterns — 8 methods, 3967 papers with code

See all 48 categories

Reinforcement Learning

- Policy Gradient Methods — 24 methods, 1717 papers with code
- Off-Policy TD Control — 17 methods, 1956 papers with code
- Reinforcement Learning Frameworks — 13 methods, 303 papers with code
- Q-Learning Networks — 11 methods, 636 papers with code
- Heuristic Search Algorithms — 9 methods, 432 papers with code

See all 28 categories

Audio

- Generative Audio Models — 17 methods, 393 papers with code
- Audio Model Blocks — 14 methods, 79 papers with code
- Text-to-Speech Models — 12 methods, 142 papers with code
- Speech Separation Models — 6 methods, 46 papers with code
- Speech Recognition — 4 methods, 43 papers with code

See all 13 categories

Sequential

- Recurrent Neural Networks — 41 methods, 9435 papers with code
- Sequence To Sequence Models — 22 methods, 2928 papers with code
- Time Series Analysis — 12 methods, 907 papers with code
- Temporal Convolutions — 8 methods, 276 papers with code
- Bidirectional Recurrent Neural Networks — 6 methods, 767 papers with code

See all 11 categories

Graphs

- Graph Models — 66 methods, 2417 papers with code
- Graph Embeddings — 44 methods, 1267 papers with code
- Graph Representation Learning — 17 methods, 9144 papers with code
- Graph Data Augmentation — 2 methods, 1 papers with code

Want to be one of the best -> master the basics

- do assignments, projects, particepate … hard work pays off

SPARK

Thanks

Deeplearning before attending SPARK

Deeplearning after attending SPARK